

Lab: Exploring Local Storage

Estimated time needed: 15 minutes

Welcome to this lab, where you will learn how to access and manipulate local storage in web browsers, with a focus on Google Chrome. Local storage is a handy feature for saving key-value pairs without expiring with the session, making it useful for preserving user preferences, settings, and other data across sessions.

Objectives

After completing this lab, you will be able to:

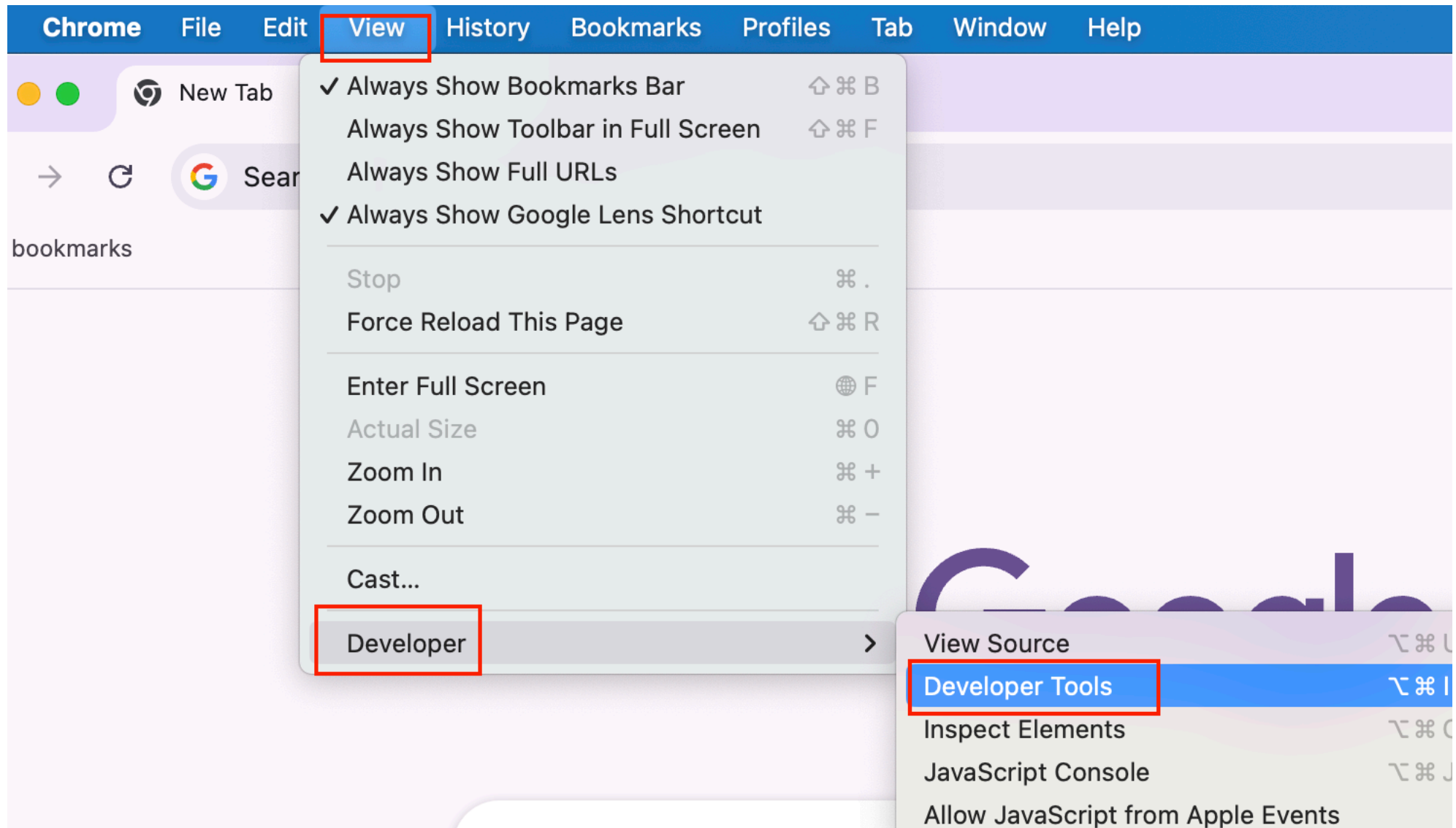
- Access local storage in different web browsers, focusing primarily on Google Chrome
- Inspect and understand the features available in local storage using developer tools
- Add, modify, and delete data in local storage
- Interact with local storage using the JavaScript console

Exercise 1: Accessing local storage

This exercise will teach you how to access local storage in Google Chrome and other popular web browsers.

1. Open Developer Tools in Chrome

- Right-click anywhere on a web page, select *Inspect*, or use the shortcut `Ctrl+Shift+I` (Windows/Linux) or `Cmd+Option+I` (Mac)
- Navigate to the *Application* tab to access the local storage section under the *Storage* panel on the left sidebar



2. Access local storage in other browsers

- **Firefox:** Open Menu > Web Developer > Storage Inspector or use the shortcut Ctrl+Shift+I and select the Storage tab.
- **Safari:** Enable the Develop menu via Safari's Preferences, then select Develop > Show Web Inspector and go to the Storage tab.

 Elements Console Sources Network **Application** >>  104  1   

Application

-  Manifest
-  Service workers
-  Storage

Storage

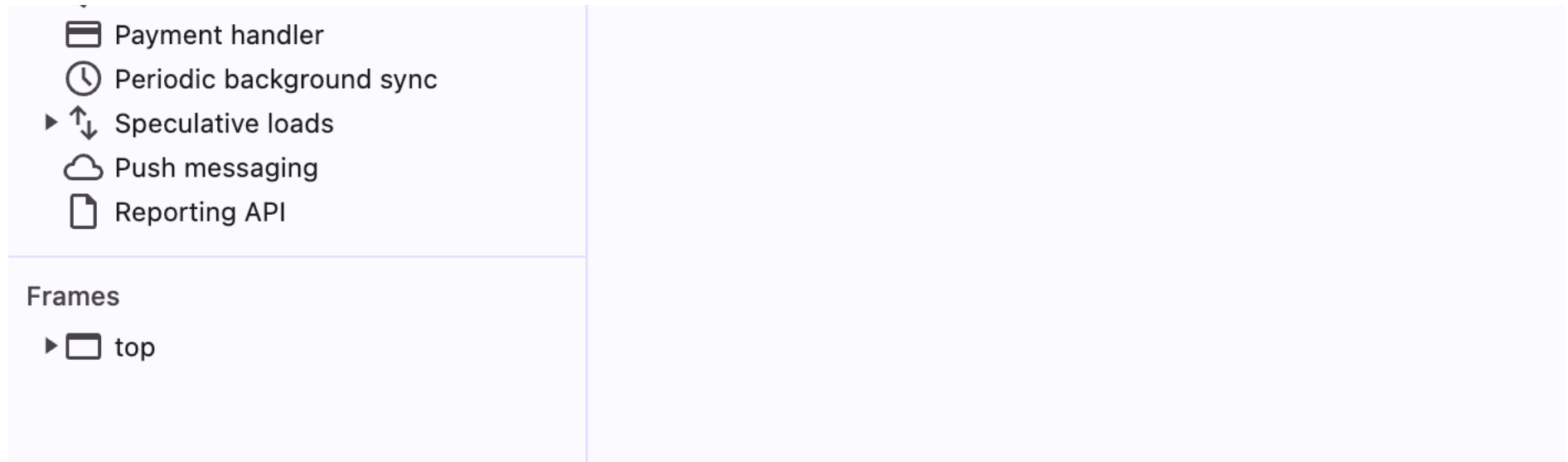
- ▼  Local storage
 -  chrome-untrusted://new-tab-page
 -  chrome://new-tab-page
 -  https://ogs.google.com
- ▶  Session storage
-  IndexedDB
- ▶  Cookies
-  Private state tokens
-  Interest groups
- ▶  Shared storage
-  Cache storage
-  Storage buckets

Background services

-  Back/forward cache
-  Background fetch
-  Background sync
-  Bounce tracking mitigations
-  Notifications

Local storage

[Learn more](#)



Exercise 2: Exploring local storage

In this exercise, you will explore local storage by inspecting Wikipedia's homepage.

1. Visit Wikipedia

- Go to <https://www.wikipedia.org/> and open Developer Tools
 - Navigate to the Application tab and select Local Storage on the left-hand menu. You should see a list of key-value pairs currently stored by the browser for Wikipedia.

2. Inspect local storage values

- Observe the entries in local storage. You might see keys like `portal_session_id` and `translationHash`, among others.

WIKIPEDIA

The Free Encyclopedia

English

6,873,000+ articles

日本語

1,427,000+ 記事

Русский

1 996 000+ статей

Español

1.974.000+ artículos

Deutsch

2.937.000+ Artikel

Français

2 631 000+ articles

Italiano

1.879.000+ voci

中文

1,438,000+ 条目 / 條目

Português

1.132.000+ artigos

فارسی

۱۰۰۱۱۰۰۰+ مقاله



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Elements

Cons

Application

-  Manifest
-  Service workers
-  Storage

Storage

- ▼  Local storage
 -  <https://www.wikipedia.org>
- ▶  Session storage
-  IndexedDB
- ▶  Cookies
-  Private state tokens
-  Interest groups
- ▶  Shared storage
-  Cache storage
-  Storage buckets

Background services

-  Back/forward cache
-  Background fetch
-  Background sync
-  Bounce tracking mitigation
-  Notifications
-  Payment handler
-  Periodic background sync

Read Wikipedia in your language



Exercise 3: Manipulating local storage

In this exercise, you will learn how to add, modify, and delete data in local storage using Developer Tools.

1. Add new key-value pairs
- In the Local Storage section of the Application tab, double-click an empty row in the table to add a new key-value pair.
 - Enter a key name, for example, testKey and a value testValue.
 - Press Enter to save the new key-value pair.

Speculative loads

Push messaging

Reporting API

Frames

top

The screenshot shows the Application tab in Chrome DevTools. The left sidebar lists the Application and Storage sections. Under Storage, 'Local storage' is expanded, and 'https://www.wikipedia.org' is selected. The main pane shows a table of key-value pairs for this origin. The entry with key 'testKey' and value 'testValue' is highlighted with a red box.

Key	Value
portal_session_id	77529f04109b20d9
portal_test_group_expires	1727235998281
translationHash	7e64cfc7
testKey	testValue

2. Modify existing key-value pairs

- Double-click the value of an existing entry. Change the value and press Enter to update.

3. Delete key-value pairs

- To delete an entry, right-click a key and select Delete.

The screenshot shows the Chrome DevTools Application tab. On the left, the 'Storage' section is expanded, showing 'Local storage' for the origin 'https://www.wikipedia.org'. A table of stored items is displayed:

Key	Value
portal_session_id	77529f04109b20d9
portal_test_group_expires	1727235998281
translationHash	7e64cfc7
testKey	

A context menu is open over the 'testKey' row, with options: 'Sort By', 'Header Options', 'Refresh', 'Edit "Key"', and 'Delete'. The 'Delete' option is highlighted with a red rectangle.

Exercise 4: Using the Console to interact with local storage

This exercise demonstrates how to use the JavaScript Console to interact with local storage programmatically.

1. Open the Console tab:

- Switch to the Console tab in Developer Tools.

2. Add a value:

- Type `localStorage.setItem('newKey', 'newValue');` and press Enter.

3. Retrieve a value:

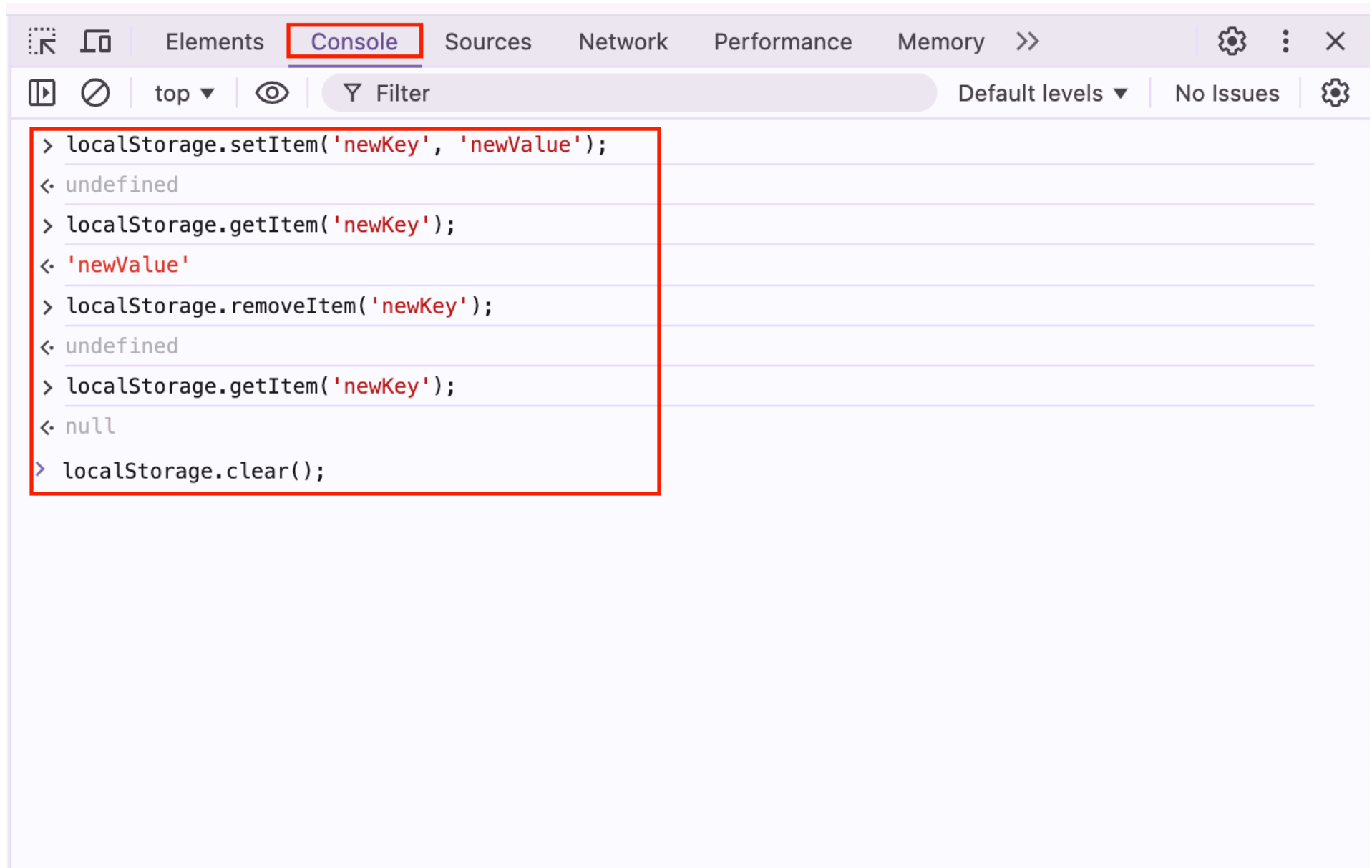
- To retrieve the value you just set, type `localStorage.getItem('newKey');` and press Enter.

4. Remove a value:

- To remove the key you added, type `localStorage.removeItem('newKey');` and press Enter.

5. Clear all local storage:

- If you want to clear all local storage entries for the site, type `localStorage.clear();` and press Enter.



Conclusion

In this lab, you have learned how to access and manipulate local storage in various web browsers, focusing on Google Chrome. You've explored how to manually add, retrieve, modify, and delete local storage data through the Developer Tools and programmatically using the Console.

By understanding local storage, you can enhance the user experience on your websites by persistently storing user preferences and other data across browser sessions.

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